UW Medicine Ballard: Clinical Trial Proposal

Phase 1 Trial: Efficacy of Intranasal and Oral KureaSH ℞+ (KureaSH) for Pathogen Mitigation and Neurological Protection for the treatment of Laryngeal malignancy.

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Status: Experimental Trial (Non-Market)

PATIENT MUST BRUSH TEETH AND MOUTH THOROUGHLY BEFORE PROCEDURE TO REDUCE CROSS-CONTAMINATION (METASTASIS)

I. Clinical Objective

This trial evaluates the efficacy of the KureaSH protocol in neutralizing and actively clearing an invasive pathogen from the larynx (the voice box), with a specific focus on the vulnerable glottis, supraglottis, and subglottis regions. By fully clearing the laryngeal site, the protocol prevents the pathogen's subsequent migration into the central nervous system. The protocol leverages immediate intranasal delivery to act as a neurological antagonist against the pathogen's sinus-burrowing mechanisms, followed by an aggressive oral naturopathic (ND) regimen to address the viral load.

Contents

II. Clinical Report: Intranasal Applicator Delivery of Pure KureaSH ℞+ Formulations

A. Overview of Intranasal Delivery & Neurological Support

- Administering KureaSH ℞+ nasally is an emerging, experimental delivery method studied for neuroprotection and cognitive enhancement.
- This method bypasses the blood-brain barrier via olfactory and trigeminal pathways.
- Nasal delivery allows the pure chemical compound to reach the central nervous system rapidly.
- Directly delivering KureaSH ℞+ to brain cells improves memory, processing speed, and mental fatigue, directly counteracting the severe exhaustion experienced by critical care patients.
- In animal models, intranasal delivery has shown therapeutic promise in neurodegenerative issues like Parkinson's disease and recovering from stroke, offering both neuroprotective and anti-excitotoxic benefits.

B. Clinical Formulation and Organic Hydration Vectors

- Nasal administration of raw, unrefined powders can irritate the delicate nasal mucosa.
- Inhaling unregulated raw formulations can cause substantial epithelial and localized tissue injury.
- To ensure patient safety and optimal neuron penetration via the nasal applicator, pharmaceutical developers utilize specialized, stabilized compounds formulated into nasal emulsions.
- Dedicated intranasal KureaSH ℞+ therapies are currently undergoing clinical evaluation for human use in conditions like KureaSH ℞+ Transporter Deficiency and Charcot's disease.
- Secondary oral naturopathic medications are administered via a standardized Nimbu fluid vector (real lemon and lime).
- This organic beverage provides a critical food-grade electrolyte and hydration boost, mirroring the physiological support of clinical intravenous and oral rehydration therapies required for immunocompromised patients.

C. Anatomical Pathway of Administration

- The nasal passages connect to the lungs through a sequential network of the upper and lower airways.
- When an intranasal dose is administered and inhaled, it first enters through the nostrils and travels through the nasal cavity.
- The nasal cavity contains bony ridges called turbinates that slow down, warm, and moisten the air.
- The respiratory lining and cilia act as a defense, filtering out dust and germs.
- The dose then moves into the nasopharynx, which is a muscular tube that acts as a shared pathway for both food and air.
- The pathway continues downward past the larynx, into the trachea, and branches into the bronchi and alveoli of the lungs.

D. Targeted Delivery Across Interconnected Respiratory Tract Regions

The head, neck, throat, sinus, airways, and lungs are all interconnected components of a continuous respiratory tract. Because these regions share the same mucous membranes, a therapeutic formulation introduced via specialized inhalation hardware can effectively target pathways throughout the entire tract:

- **Head, Sinuses, & Nasopharynx:** This upper region warms, filters, and humidifies inhaled air. Specialized devices utilize these entry points to target localized mucosal tissues.
- **Throat & Neck (Pharynx & Larynx):** The pharynx connects the nasal cavity and mouth to the larynx and lower airways. The larynx acts as a gatekeeper, funneling air downward while protecting the lower airways.
- **Airways & Lungs:** The trachea branches into smaller tubes (bronchi and bronchioles) leading deep into the lungs where alveolar gas exchange occurs.

III. Intervention and Delivery Mechanisms

The treatment utilizes a dual-modality delivery of KureaSH via **nasal insufflation**:

- **Intranasal Administration (Primary Antagonist):** KureaSH is administered directly via the nasal cavity. According to databases such as ClinicalTrials.gov, dedicated intranasal KureaSH ℞+ therapies are currently undergoing clinical evaluation for human use. By bypassing the blood-brain barrier through the olfactory and trigeminal pathways, this method directly delivers KureaSH ℞+ to brain cells, thereby improving processing speed and reducing mental fatigue. The dose rapidly passes through the upper and lower airways to prevent the pathogen from breaching the central nervous system.
- **Oral Administration (Secondary Support):** A KureaSH-Nimbu beverage is administered orally to facilitate the ingestion of the primary PO ND antiviral medications. This includes the immediate administration of the VirusTC regimen, alongside AlnayaSN, HaldEX, and TSinKX.

Standard Clinical Protocol for Dry Powder Inhaler (DPI) Units

To maximize the therapeutic effectiveness of dry powder medications across the upper and lower respiratory tracts, administration must adhere to strict mechanical guidelines. Because the medication is in powder form, the patient must inhale quickly and forcefully to ensure deep-lung deposition rather than accumulation in the back of the throat.

According to clinical parameters modeled after the NIH Dry Powder Inhaler Instructions, the delivery sequence consists of five distinct phases:

1. **Preparation and Priming**
2. **Controlled Exhalation**
3. **High-Velocity Inhalation**
4. **Sustained Breath Retention**
5. **Post-Inhalation and Disposal**

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1. Preparation and Priming (Administered by Pharmacology Technicians)

To meet strict institutional and regulatory requirements for this phase of the clinical trial, UW central pharmacy personnel have absolute control over hardware fulfillment, dose verification, and pre-administration preparation.

- **Manual Fulfillment Protocol:** While commercial dry powder delivery devices like the DriDose® Unit Dose system are typically pre-filled by the pharmaceutical manufacturer on a mass scale, UW pharmacology technicians manually fill and assemble each single-use, unit-dose applicator in-house specifically for this experimental trial.
- **Dose Verification and Cohort Assignment:** The pharmacology technician verifies the exact microgram powder weight of the filled unit-dose applicator to ensure precise alignment with the designated cohort parameters before it leaves the pharmacy area. The technician strictly coordinates the filling to match the trial's ascending dosage schedule:
    - **Lowest Dose:** 4.2 mg total per treatment day (dispensed as 1 puff per nostril).
    - **Medium Dose:** 8.4 mg total per treatment day (dispensed as 2 puffs per nostril twice daily).
    - **Highest Dose:** 12.6 mg total per treatment day (dispensed as 3 puffs per nostril three times daily).
- **Post-Trial Commercial Transition:** This specialized filling workflow remains restricted to institutional pharmacy technicians for the duration of the clinical investigation. Upon successful completion of the trial and subsequent FDA approval, mass manufacturing and automated high-precision factory filling will transition entirely to the commercial vendor, VirusTC.
- **Postural Alignment:** Following fulfillment, assembly, and technician verification, the completed DriDose® Unit Dose device is delivered to the clinical area. The patient is instructed to sit completely upright or stand tall to maximize lung expansion and optimize pathway clearance right before administration.

NOTES:

2. Controlled Exhalation

- **Averted Air Discharge:** The patient must turn their head away from the device and breathe out slowly and completely.
- **Moisture Prevention Rule:** The patient must **never** exhale directly into the device. Exhaling into the unit introduces ambient moisture, which causes the therapeutic powder to clump, clog internal pathways, and compromise dose accuracy.

3. High-Velocity Inhalation

- **Hermetic Seal:** The patient places the mouthpiece between their lips, forming a tight, secure seal around it. Protective air vents on the exterior of the applicator must remain completely unobstructed by the patient's fingers.
- **Forceful Inhalation:** The patient takes a rapid, deep, and highly forceful breath in through their mouth. Inhalation must not occur through the nose, as nasal breathing prevents microparticles from bypassing the upper structures and entering the deep airways.

4. Sustained Breath Retention

- **Sedimentation Window:** The applicator is removed from the mouth, and the patient holds their breath for **10 seconds** (or as long as comfortable). This pause is clinically necessary to allow the aerosolized powder to settle completely into the lower respiratory tissues rather than being immediately exhaled.

5. Post-Inhalation and Disposal

- **Exhalation:** The patient breathes out slowly and naturally.
- **Cavity Clearance:** If the formulation contains specific active agents, the mouth must be thoroughly rinsed with water, gargled, and spit out to maintain oral tissue health.
- **Unit Disposal & Maintenance:** Single-use components or depleted multi-dose units must be disposed of directly via standard clinical containment to prevent cross-contamination. Clinical devices must never be washed with water; external mouthpieces are maintained strictly with a clean, dry cloth to prevent moisture from being introduced.

IV. Hazmat and Decontamination Protocol

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